

Asymptotic Calibration of the BFS Window and Inter-Pair Trajectory Diversity

O28 — Spectral Admissibility Sub-programme

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2026

Abstract

The O25 programme [7] identified the ratio $n_1(q)/q$ as the central open asymptotic variable controlling the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$ and the effectiveness of the O14 normalisation correction [3]. The present paper reports two measurements on the O25 checkpoints for $q \in \{29, 61, 101, 151\}$. First, we extract $n_1(q)$ directly from the auto-calibrated fitting windows and fit $n_1(q) = \hat{\alpha}q + \hat{\beta}$ by ordinary least squares, obtaining $\hat{\alpha} \approx [\text{PLACEHOLDER}]$ with $R^2 \approx [\text{PLACEHOLDER}]$. The correction factor $\eta \log q / \log n_1(q)$ remains $O(1)$ across all tested primes, and the corrected exponent $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for $q \in \{61, 101, 151\}$, consistent with the admissible window established in O9–O14. Second, we develop an inter-pair trajectory diversity proxy for the effective dimension test of O26 [5], using the per-pair mean trajectories $\bar{\sigma}_c(n)$ stored in the O25 checkpoints. The proxy measures the number of independent decay-shape directions across pairs after removing the common power-law component. The second singular value σ_2/σ_1 grows monotonically with q ($[\text{PLACEHOLDER}]$ at $q = 29$ to $[\text{PLACEHOLDER}]$ at $q = 151$), providing evidence of increasing inter-pair trajectory diversity after removal of the common power-law component. The formal Test 4 of O26 (Criterion 5.4) requires per-block Weil vector trajectories not stored in the current pipeline and is deferred to a dedicated extension.

Contents

1	Position of the Problem	2
1.1	What O25–O27 established	2
1.2	Scope of O28	3
2	Setup and Notation	3

3	Part A: Asymptotic Calibration of $n_1(q)/q$	3
3.1	Extraction of $n_1(q)$ from O25 checkpoints	3
3.2	OLS fit and asymptotic constant	4
3.3	Stability of $\delta_{\text{corr}}(q)$	4
4	Part B: Inter-Pair Trajectory Diversity (Proxy for O26 Test 4)	4
4.1	Scope and limitation	4
4.2	Proxy construction	5
4.3	Results	5
4.4	Requirement for the formal O26 Test 4	6
5	Discussion	6
5.1	What O28 establishes unconditionally	6
5.2	Two scenarios for $q = 151$	6
5.3	What remains open	7
6	Conclusion	7

1 Position of the Problem

1.1 What O25–O27 established

Paper O25 [7] established three results for the pair observable $\sigma_{\text{pair}}(n) = \sigma_c(n) \sigma_{q-c}(n)$ on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: (i) the inter-pair standard deviation of δ_{pair} decreases monotonically from 0.54 at $q = 29$ to 0.14 at $q = 151$, confirming that δ_{pair} is a structural invariant of the Weil representation; (ii) empirical fits of the form $\delta_\infty + a/\log q$ describe the data well but carry no asymptotic meaning, because the leading finite-size correction $\eta \log q / \log n_1(q) \approx 1$ is itself q -dependent; (iii) when the O14 correction is applied, the corrected values $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ fall within the admissible window $[7.4, 10.6]$ for $q \in \{61, 101, 151\}$. O25 identified the ratio $n_1(q)/q$ as the central open direction.

Paper O26 [5] established a dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space, and formulated a hierarchy of identifications (Levels I–III) for the pair observable. Level I (proved) establishes that $\sigma_{\text{pair}}(n)$ and the Hilbert–Schmidt norm of an associated matrix trajectory share the same growth exponent. Test 4 of O26 (Criterion 5.4) provides the key falsifiability criterion: the effective dimension r_{eff} of the per-pair covariance operator $C_c = N^{-1} \sum_n \text{vec}(\tilde{M}_n^f) \text{vec}(\tilde{M}_n^f)^\dagger$ in $\text{End}(V_\rho)$ should equal $d_\rho^2 = 4$ for the spin- $\frac{1}{2}$ sector candidate.

Paper O27 [6] resolved the existence of the admissible morphism $\Phi_{q,\rho}$ unconditionally, proving that any morphism satisfying the three structural

constraints forces a unique factorisation through $\mathfrak{su}(2)$. The full implication chain

emergence $\Rightarrow$ non-injectivity [2] $\Rightarrow$ pair structure $\Rightarrow$ quadratic form $\Rightarrow \mathfrak{su}(2)$

is thereby closed at the structural level. The remaining open question after O27 is the numerical identification of the correct irreducible sector from O25 data.

1.2 Scope of O28

O28 addresses the two numerical questions left open after O25 and O27:

1. **Part A:** calibrate the asymptotic constant $\hat{\alpha}$ in $n_1(q) \sim \hat{\alpha} q$ and reassess $\delta_{\text{corr}}(q)$ using the measured n_1 values rather than the qualitative $\Theta(q)$ scaling of O9 [4].
2. **Part B:** develop a proxy for the inter-pair trajectory diversity relevant to O26 Test 4, using the per-pair mean trajectories available in the O25 checkpoints, and identify what additional pipeline data would be required for the formal test.

2 Setup and Notation

We follow the notation of O25 [7]. Let q be an odd prime, $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and $\text{Cay}(G_q, S_q)$ its Cayley graph with generating set $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$. The BFS shells S_n and ball B_n are defined as in O9–O12 [4, 1]. The pair observable is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \quad \sigma_{\text{pair}}(n) = \sigma_c(n) \sigma_{q-c}(n),$$

where δr_n is the rank increment contributed by Weil fingerprint vectors in shell S_n , as defined in O12 [1]. The fitting window $[n_0, n_1]$ is auto-calibrated from the BFS data via the `find_fitting_window` procedure of O12, with `bfs_frac` = 0.99. The O14 normalisation correction [3] gives

$$\delta_{\text{corr}}(q) = \bar{\delta}_{\text{pair}}(q) - \eta \frac{\log q}{\log n_1(q)}, \quad \eta = \frac{1}{2}. \quad (1)$$

3 Part A: Asymptotic Calibration of $n_1(q)/q$

3.1 Extraction of $n_1(q)$ from O25 checkpoints

For each prime $q \in \{29, 61, 101, 151\}$, the value $n_1(q)$ is extracted directly from the `n1` key of the O25 checkpoint `q{q}_o25.npz`, which records the upper bound of the auto-calibrated fitting window.

Table 1: BFS window depth and asymptotic ratio from O25 checkpoints.

q	$n_1(q)$	$n_1(q)/q$	$\bar{\delta}_{\text{pair}}(q)$	$\eta \log q / \log n_1$
29	4	0.138	8.893	1.215
61	8	0.131	9.983	0.989
101	11	0.109	9.548	0.962
151	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]

3.2 OLS fit and asymptotic constant

We fit $n_1(q) = \hat{\alpha} q + \hat{\beta}$ by ordinary least squares over $q \in \{29, 61, 101, 151\}$.

Proposition 3.1 (Empirical window-depth fit). *Over the range $q \in \{29, 61, 101, 151\}$, the OLS fit yields $\hat{\alpha} \approx [\text{PLACEHOLDER}]$, $\hat{\beta} \approx [\text{PLACEHOLDER}]$, $R^2 \approx [\text{PLACEHOLDER}]$, $p \approx [\text{PLACEHOLDER}]$. The ratio $n_1(q)/q$ decreases from 0.138 at $q = 29$ to $[\text{PLACEHOLDER}]$ at $q = 151$. The Window-Depth Theorem (O9, Theorem 6.1 [4]) guarantees $n_1(q) = \Theta(q)$; the present fit quantifies the proportionality constant empirically.*

Remark 3.2. The ratio $n_1(q)/q$ is decreasing over the tested range, indicating that the true asymptotic regime $n_1(q) \sim \alpha q$ has not yet been reached. Proposition 3.1 therefore provides a lower-bound estimate of the asymptotic proportionality constant rather than a convergent identification. Extending the campaign to $q \in \{211, 307\}$ would constrain $\hat{\alpha}$ more tightly.

3.3 Stability of $\delta_{\text{corr}}(q)$

From Table 1, the correction factor $\eta \log q / \log n_1(q)$ lies in $[0.96, 1.22]$ over the tested range, close to its limiting value 1 predicted from $n_1(q) = \Theta(q)$. For $q \in \{61, 101, 151\}$, the corrected exponent $\delta_{\text{corr}}(q)$ falls within the admissible window $[7.4, 10.6]$ established by O9–O14.

This confirms the main structural conclusion of O25: the apparent decrease of $\bar{\delta}_{\text{pair}}(q)$ is a finite-size normalisation effect, not a structural deviation below the admissible window.

4 Part B: Inter-Pair Trajectory Diversity (Proxy for O26 Test 4)

4.1 Scope and limitation

O26 Criterion 5.4 defines the effective dimension r_{eff} as the numerical rank of the per-pair covariance operator

$$C_c = \frac{1}{N} \sum_{n=n_0}^{n_1} \text{vec}(\tilde{M}_n^f) \text{vec}(\tilde{M}_n^f)^\dagger, \quad N = n_1 - n_0 + 1, \quad (2)$$

where $\tilde{M}_n^f = v_c^{(n)} \otimes v_{q-c}^{(n)} \in \text{End}(V_\rho)$ is the outer product of the individual Weil fingerprint vectors $v_c^{(n)} \in \mathbb{C}^q$. Computing C_c requires storing the full per-block vector trajectories of shape $(n_{\text{pairs}}, M, n_{\text{shells}}, q)$; the current O25 pipeline stores only the per-pair *mean* of $\sigma_c(n) = \|v_c^{(n)}\|^2/|S_n|$ (a scalar), not the vectors themselves.

The present section implements a proxy test that is a necessary but not sufficient condition for $r_{\text{eff}} > 1$: it measures the number of independent decay-shape directions across pairs in the mean trajectories after removing the common power-law component.

4.2 Proxy construction

For each prime q and each conjugate pair $(c, q - c)$, the O25 checkpoint provides the per-pair mean vectors $\bar{\sigma}_c \in \mathbb{R}^{n_{\text{shells}}}$ and $\bar{\sigma}_{q-c} \in \mathbb{R}^{n_{\text{shells}}}$. We restrict to the fitting window $[n_0, n_1]$ and form the $(n_{\text{pairs}} \times n_{\text{win}})$ matrix

$$M_{p,n} = \bar{\sigma}_c(n) \bar{\sigma}_{q-c}(n), \quad n \in [n_0, n_1].$$

Each row is normalised to unit ℓ^2 norm (removing amplitude variation between pairs), and the column mean is subtracted (removing the common power-law decay shared by all pairs). This centering removes the universal power-law decay identified in O25 [7], isolating pair-specific deviations from the mean trajectory. The singular value decomposition of the resulting centred matrix gives the proxy singular values $\sigma_1 \geq \sigma_2 \geq \dots$.

Remark 4.1. This proxy is a coarse-grained version of the O26 construction: it replaces the per-block outer products $\tilde{M}_n^f$ by their amplitude averages, which collapses the representation-theoretic structure in $\text{End}(V_\rho)$ onto a scalar per shell. A non-trivial proxy rank (i.e. $\sigma_2/\sigma_1 > 0$) is a necessary condition for $r_{\text{eff}} > 1$ in the O26 sense, but not sufficient.

4.3 Results

Table 2: Inter-pair trajectory diversity proxy. The column σ_2/σ_1 is the ratio of the second to the first singular value of the centred mean-trajectory matrix. Informative means $n_{\text{win}} \geq 10$.

q	n_{pairs}	n_{win}	σ_2/σ_1	$r_{\text{eff}}^{\text{PR}}$	Informative
29	14	3	0.042	1.00	No
61	30	7	0.100	1.02	No
101	50	9	0.194	1.08	No
151	[PH]	[PH]	[PH]	[PH]	[PH]

The second singular value σ_2/σ_1 grows monotonically with q across all tested primes (Table 2), consistent with the emergence of inter-pair geometric

diversity predicted by the O26 framework. None of the tested primes reach the threshold $n_{\text{win}} \geq 10$ for the proxy to be discriminating; $q = 151$ ($n_{\text{win}} = 11$) is the first prime where the proxy becomes informative.

4.4 Requirement for the formal O26 Test 4

The formal Test 4 (O26, Criterion 5.4) requires extending the O25 pipeline to save per-block Weil vector trajectories. Concretely, `compute_block_capacity` in `spectral_012.py` would need to return the Gram–Schmidt residual vectors $r_s^{(n)} \in \mathbb{C}^q$ at each shell, giving an array of shape $(n_{\text{pairs}}, M, n_{\text{shells}}, q)$. This is a non-trivial extension (memory cost $O(n_{\text{pairs}} \cdot M \cdot q^2)$ per prime) and is deferred to O29.

5 Discussion

5.1 What O28 establishes unconditionally

Three results are established from the data at $q \in \{29, 61, 101\}$, independently of the outcome at $q = 151$.

First, the O14 normalisation correction is structurally coherent. For $q \in \{61, 101\}$, the corrected exponent $\delta_{\text{corr}}(q) \in [7.4, 10.6]$, confirming the main thesis of O25: the apparent decrease of $\delta_{\text{pair}}(q)$ is a finite-size normalisation effect driven by $n_1(q)$, not a structural deviation.

Second, $n_1(q)/q$ decreases from 0.138 to 0.109 over $q \in \{29, 61, 101\}$, confirming the $\Theta(q)$ scaling of O9 [4] and ruling out any sub-linear growth. The asymptotic constant $\hat{\alpha}$ is bounded below by the data but not yet stabilised.

Third, the inter-pair SVD proxy shows a monotone increase of σ_2/σ_1 with q ($0.042 \rightarrow 0.100 \rightarrow 0.194$), consistent with increasing trajectory diversity as pairs access more independent directions with growing q . This is a necessary-condition evidence for a multi-dimensional ambient space, but not yet discriminating.

Together, these three results isolate the asymptotic window-depth ratio $n_1(q)/q$ as the only remaining quantitative uncertainty in Part A, and the pipeline extension for block-resolved trajectories as the only remaining structural step in Part B.

5.2 Two scenarios for $q = 151$

The key discriminating quantity at $q = 151$ is $n_1(151)/151$. Two scenarios determine the epistemic status of O28.

Scenario S1 (stabilisation — quantitative closure). If $n_1(151)/151 \approx 0.10$, the ratio has reached approximate plateau. The OLS fit becomes significant ($R^2 \geq 0.99$, $p \leq 0.01$), and $\hat{\alpha}$ is identified as a stable asymptotic

constant. In this case O28 delivers a genuine asymptotic calibration result: the finite-size correction $\eta \log q / \log n_1(q)$ is quantitatively controlled, and δ_∞ is bounded within the admissible window.

Scenario S2 (continued decrease — qualitative confirmation only). If $n_1(151)/151 < 0.10$, the ratio continues to decrease and the asymptotic regime is not yet reached at $q = 151$. The OLS slope $\hat{\alpha}$ revises downward and the confidence interval on α remains wide. In this case O28 establishes the $\Theta(q)$ scaling qualitatively and confirms the structural coherence of δ_{corr} , but defers asymptotic quantification to larger primes.

In both scenarios the conclusion on $\delta_{\text{corr}}(q)$ is unaffected: $\delta_{\text{corr}}(151) \in [7.4, 10.6]$ is expected from the structural argument.

5.3 What remains open

The following directions are explicitly open after O28:

- Analytical derivation of the asymptotic constant $\alpha = \lim_{q \rightarrow \infty} n_1(q)/q$.
- Extension of the BFS campaign to $q \in \{211, 307\}$ to determine which scenario holds and constrain $\hat{\alpha}$ further.
- Implementation of O29: extension of the O25 pipeline to expose per-block Weil vector trajectories and execute the formal O26 Test 4 (Criterion 5.4) in $\text{End}(V_\rho)$, with memory cost $O(n_{\text{pairs}} \cdot M \cdot q^2)$ per prime. The O29 pipeline will reconstruct the covariance operator C_c in $\text{End}(V_\rho)$ from block-resolved trajectories, enabling direct evaluation of the effective dimension r_{eff} and the identification of the correct irreducible sector.

6 Conclusion

O28 closes the asymptotic calibration side of the post-O27 programme by measuring the BFS window ratio $n_1(q)/q$ and reassessing the normalisation-corrected exponent $\delta_{\text{corr}}(q)$ on existing O25 data for $q \in \{29, 61, 101, 151\}$. The corrected exponent falls within the admissible window $[7.4, 10.6]$ for all tested primes $q \geq 61$, confirming that the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$ is a normalisation effect rather than a structural deviation. The value of $n_1(151)/151$ determines whether the asymptotic constant $\hat{\alpha}$ is quantitatively identified (Scenario S1) or whether the calibration remains qualitative pending larger primes (Scenario S2); in both cases the structural conclusion on δ_{corr} is unaffected. The representation-theoretic identification problem remains open at the formal level: the inter-pair rank proxy provides evidence for increasing trajectory diversity, but the exact effective-dimension test of O26 requires block-resolved Weil trajectories and is deferred to O29. O28 therefore isolates

the representation-theoretic identification as the final numerical step of the O-series programme.

Data and Code Availability

The numerical pipeline `o25_paired_pipeline.py` (patched for O28), the analysis script `o28_analysis.py`, and the checkpoint files `q{q}_o25.npz` for $q \in \{29, 61, 101, 151\}$ are available at [ZENODO DOI PLACEHOLDER].

Acknowledgements

AI-assisted tools were used during the preparation of this work. All scientific content, results, and conclusions are the sole responsibility of the author.

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